Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9				**Indicative content** Observations • sodium iodide turns brown with both chlorine and bromine • sodium bromide turns orange with chlorine • no reaction when iodine is added to sodium chloride or sodium bromide or when bromine is added to sodium chloride Conclusions • chlorine displaces both bromine and iodine from bromide/iodide solutions • chlorine is therefore most reactive • bromine displaces iodine from iodide solution and is therefore more reactive than iodine • more reactive halogens displace less reactive halogens from solution trend in reactivity - chlorine > bromine > iodine Equations • $Cl_2 + 2NaBr \rightarrow 2NaCl + Br_2$ • $Cl_2 + 2NaI \rightarrow 2NaCl + I_2$ • $Br_2 + 2NaI \rightarrow 2NaBr + I_2$	4	2		6		4
				5-6 marks Accurate observations and conclusions; good attempt at two equations *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3-4 marks** Two observations and partial conclusion; attempt at one equation *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** One observation and attempt at conclusion *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
				Question 9 total	**4**	**2**	**0**	**6**	**0**	**4**